

Certified higher theta moments and the original gamma arithmetic coefficients

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This continuation proves and evaluates the fourth and sixth moments of the original theta seed, then transports the enclosures through the exact coefficient map for the reference parameter $\lambda = 13/4$ in *An exact gamma pushforward for the arithmetic Toda control*. The seed retains $g = 2\xi$, the density $|g(1/2 + it)|^2/(2\pi)$, the Fourier coordinate t , and its actual total mass. The reference retains the argument $\lambda + it/2$ and its different actual mass. We give the full analytic formulas, an explicit infinite-square tail for every finite derivative order, and the exact rational enclosures used by the executable certificate.

1 The theta input and its transform, with the original factors

On $\mathcal{S} = \{z \neq 0 : |\arg z| < \pi/4\}$ use the principal logarithm, and put

$$D = -z\partial_z, \quad L = D - \frac{1}{2}, \quad \vartheta(z) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 z^2}, \quad (\text{GS.1})$$

$$f_0(z) = D(D-1)\vartheta(z) = 2 \sum_{n \geq 1} (4\pi^2 n^4 z^4 - 6\pi n^2 z^2) e^{-\pi n^2 z^2}. \quad (\text{GS.2})$$

The series and each differentiated series converge locally uniformly: on a compact subset of $\mathcal{S}$, every summand is bounded by a polynomial in n times e^{-an^2} for some $a > 0$.

Lemma 1 (Inversion and both endpoints). *On the specified sector,*

$$\vartheta(z) = z^{-1}\vartheta(1/z), \quad f_0(z) = z^{-1}f_0(1/z), \quad f_0(\bar{z}) = \overline{f_0(z)}. \quad (\text{GS.3})$$

Every D -derivative of f_0 decays faster than every power both at zero and at infinity on a closed proper subsector.

Proof. For $x > 0$, Fourier transformation with kernel $e^{-2\pi i \eta u}$ maps $e^{-\pi x^2 u^2}$ to $x^{-1}e^{-\pi \eta^2/x^2}$. The value at $\eta = 0$ follows from the Gaussian integral, whose square in polar coordinates is one. Differentiation in η and integration by parts give $I'(\eta) = -2\pi \eta I(\eta)/x^2$, which proves the stated formula with that initial value. Periodize this Gaussian over integer translates. Its Fourier coefficients are the displayed rapidly decreasing transform values; its absolutely convergent Fourier series evaluated at zero proves $\vartheta(x) = x^{-1}\vartheta(1/x)$. Holomorphic uniqueness extends the identity to $\mathcal{S}$.

For $(RF)(z) = z^{-1}F(1/z)$, direct differentiation gives $DR = R(1-D)$ and hence $D(D-1)R = RD(D-1)$. Applying this to $R\vartheta = \vartheta$ proves the second equality in (GS.3). The real series coefficients prove the third. On $|\arg z| \leq \beta < \pi/4$ one has $\Re z^2 \geq |z|^2 \cos(2\beta) > 0$; the series and each derivative therefore have Gaussian decay at infinity with a fixed polynomial factor. Inversion and $DR = R(1-D)$ transport all these bounds to zero, with every derivative retained. $\square$

Define, on these rapid sector functions,

$$(\mathcal{M}F)(s) = \int_0^\infty F(x)x^{s-1} dx, \quad (UF)(y) = e^{y/2}F(e^y), \quad (\mathcal{F}_+u)(t) = \int_{\mathbb{R}} u(y)e^{ity} dy. \quad (\text{GS.4})$$

Here $U : L^2(\mathbb{R}_{>0}, dx) \rightarrow L^2(\mathbb{R}, dy)$ is a surjective linear isometry: $x = e^y$ proves the norm identity, and its inverse is $u \mapsto [x \mapsto x^{-1/2}u(\log x)]$. The endpoint bounds prove that the Mellin integral is entire. For $\Re s > 1$, absolute integration of (GS.2) is justified by $\sum_{n \geq 1} n^{-\Re s} < \infty$. Substitution $v = \pi n^2 x^2$ then gives

$$\begin{aligned} (\mathcal{M}f_0)(s) &= \pi^{-s/2} \zeta(s) \{4\Gamma(s/2 + 2) - 6\Gamma(s/2 + 1)\} \\ &= s(s-1)\pi^{-s/2} \Gamma(s/2) \zeta(s) =: g(s) = 2\xi(s). \end{aligned} \quad (\text{GS.5})$$

The factor 2 in (GS.2) and the substitution factor 1/2 both occur before the first equality. Integration by parts proves the gamma recurrence used in the second equality. The entire Mellin integral continues the right side. Substitution $x \mapsto 1/x$ and real conjugation in (GS.4) prove

$$g(1-s) = g(s), \quad g(\bar{s}) = \overline{g(s)}. \quad (\text{GS.6})$$

In particular the following original density is even:

$$w(t) = \frac{|g(1/2 + it)|^2}{2\pi}, \quad \mu_j = \int_{\mathbb{R}} t^j w(t) dt. \quad (\text{GS.7})$$

All moments exist. To see the exponential bound directly, rotate the Mellin contour to the ray $e^{i\beta}\mathbb{R}_{>0}$ with $|\beta| < \pi/4$. The endpoint bounds make the closing arcs vanish, and the rotated formula has the factor $e^{i\beta s}$. On the line $\Re s = 1/2$, choosing the sign of β to match $\Im s$ yields $|g(1/2 + it)| \leq C_\beta e^{-|\beta||t|}$. Thus $\int e^{a|t|} w(t) dt < \infty$ for every $0 < a < \pi/2$. Evenness gives $\mu_{2r+1} = 0$.

2 Every derivative polynomial and the exact higher-moment map

Let $H_r = L^r f_0$ for integers $r \geq 0$. Direct differentiation of (GS.4) gives $ULF = -(UF)'$ on the rapid sector functions. Integration by parts in $\mathcal{F}_+$, with the endpoint bounds already proved, therefore gives the commuting transform identity

$$\mathcal{F}_+(UH_r)(t) = (it)^r g(1/2 + it). \quad (\text{GS.8})$$

Every sign and power i^r belongs to this map.

Theorem 2 (Exact higher-moment formula). *Define polynomials $P_r \in \mathbb{Q}[y]$ by*

$$P_0(y) = 8y^2 - 12y, \quad P_{r+1}(y) = -2yP'_r(y) + (2y - \tfrac{1}{2})P_r(y), \quad (\text{GS.9})$$

$$P_r(y) = \sum_{a=1}^{r+2} p_{r,a} y^a, \quad p_{r+1,a} = 2p_{r,a-1} - (2a + \tfrac{1}{2})p_{r,a}, \quad (\text{GS.10})$$

where coefficients outside their indicated range are zero. Then

$$H_r(x) = \sum_{n \geq 1} P_r(\pi n^2 x^2) e^{-\pi n^2 x^2}, \quad \mu_{2r} = \int_0^\infty |H_r(x)|^2 dx = 2 \int_1^\infty H_r(x)^2 dx. \quad (\text{GS.11})$$

The first four complete polynomials are

$$\begin{aligned} P_0 &= 8y^2 - 12y, \\ P_1 &= 16y^3 - 60y^2 + 30y, \\ P_2 &= 32y^4 - 224y^3 + 330y^2 - 75y, \\ P_3 &= 64y^5 - 720y^4 + 2116y^3 - 1635y^2 + \frac{375}{2}y. \end{aligned} \tag{GS.12}$$

Proof. With $y = \pi n^2 x^2$, the operator D is $-2y\partial_y$. Consequently $L(P(y)e^{-y}) = \{-2yP'(y) + (2y - 1/2)P(y)\}e^{-y}$. This proves (GS.9), the series identity in (GS.11), and (GS.10) by collecting each specified coefficient. It also proves inductively that the degree is $r + 2$, the leading coefficient is 2^{r+3} , and the constant coefficient is zero. Applying the recurrence three times gives the complete rational lists (GS.12).

Fourier Plancherel in the convention (GS.4) has factor $1/(2\pi)$: $\int |u|^2 dy = (2\pi)^{-1} \int |\mathcal{F}_+ u|^2 dt$. For the Schwartz functions here this follows by inserting $e^{-\varepsilon t^2}$ into Fourier inversion, applying Fubini, and evaluating the Gaussian kernel of integral one; letting $\varepsilon \downarrow 0$ gives inversion and the inner-product identity. Applying it to (GS.8) and the unitary U proves the first integral equality in (GS.11).

The relation $DR = R(1 - D)$ gives $RL = -LR$, while $Rf_0 = f_0$. Thus $RH_r = (-1)^r H_r$, or $H_r(1/x) = (-1)^r x H_r(x)$ for $x > 0$. In the integral over $(0, 1)$ substitute $x = 1/v$. The factor v^2 from the squared identity cancels the Jacobian v^{-2} and yields exactly the integral over $(1, \infty)$. The sign $(-1)^r$ is retained before squaring. All values $H_r(x)$ are real because their series has real coefficients, which finishes (GS.11). $\square$

The numerical driver in the delivered Gamma note writes $q_0 = 4y^2 - 6y$ and uses the same linear recurrence for q_r . Its exact coefficient map to this presentation is $q_r \mapsto P_r = 2q_r$, with inverse $P_r \mapsto P_r/2$. Thus $H_r = 2 \sum_n q_r(\pi n^2 x^2) e^{-\pi n^2 x^2}$. At the integral level (GS.11) consequently gives a factor 8 in front of the sum using $q_r q_r I_{2a+2b}$. At the gamma-function level below it gives a factor 4 in front of $q_r q_r \Gamma/C^\alpha$. The coefficient map accounts for both presentations without changing the theta input or its mass.

3 The incomplete-gamma square and a proved omitted tail

For real $C > 0$ and integers $p \geq 0$ put

$$I_p(C) = \int_1^\infty x^p e^{-Cx^2} dx = \frac{\Gamma((p+1)/2, C)}{2C^{(p+1)/2}}, \quad \Gamma(a, C) = \int_C^\infty v^{a-1} e^{-v} dv. \tag{GS.13}$$

Substitution $v = Cx^2$ proves the equality. Only positive real arguments and their positive real powers occur here. The companion recurrence, including its endpoint term, is

$$I_{p+2}(C) = \frac{p+1}{2C} I_p(C) + \frac{e^{-C}}{2C}. \tag{GS.14}$$

It follows by integration by parts in (GS.13), whose upper endpoint vanishes exponentially.

Let $H_{r,J}$ restrict the original series (GS.11) to $1 \leq n \leq J$, where $J \geq 0$ is an integer. Put $T_{r,J} = 2 \int_1^\infty H_{r,J}^2 dx$. The entire finite expression is

$$T_{r,J} = \sum_{m,n=1}^J \sum_{a,b=1}^{r+2} p_{r,a} p_{r,b} (\pi m^2)^a (\pi n^2)^b \frac{\Gamma(a+b+1/2, \pi(m^2+n^2))}{[\pi(m^2+n^2)]^{a+b+1/2}}. \tag{GS.15}$$

Here the endpoint factor 2 in (GS.11) cancels the factor 1/2 in (GS.13); the coefficients $p_{r,a}$ already contain the complete original theta factor. Empty sums at $J = 0$ have value zero.

To specify the tail by finite positive expressions, define Stirling numbers of the second kind by $\left\{ \begin{smallmatrix} 0 \\ 0 \end{smallmatrix} \right\} = 1$, zero when one index is zero and the other positive, zero outside $0 \leq b \leq j$, and

$$\left\{ \begin{smallmatrix} j+1 \\ b \end{smallmatrix} \right\} = b \left\{ \begin{smallmatrix} j \\ b \end{smallmatrix} \right\} + \left\{ \begin{smallmatrix} j \\ b-1 \end{smallmatrix} \right\}.$$

For $a \geq 1$, $d > 0$, integers $p \geq 0$, set

$$\rho_a = e^{-d(2a+1)},$$

$$\mathbf{H}_p(a, d) = \sum_{j=0}^p \binom{p}{j} a^{p-j} \sum_{b=0}^j \left\{ \begin{smallmatrix} j \\ b \end{smallmatrix} \right\} \frac{b! \rho_a^b}{(1 - \rho_a)^{b+1}}, \quad (\text{GS.16})$$

$$\mathbf{A}_r(a) = \sum_{b=1}^{r+2} |p_{r,b}| \pi^b \mathbf{H}_{2b}(a, \pi). \quad (\text{GS.17})$$

Theorem 3 (Every finite derivative order has an explicit square tail). *For every integer $r \geq 0$ and $J \geq 0$, with $K = J + 1$,*

$$|\mu_{2r} - T_{r,J}| \leq E_{r,J} := 4\mathbf{A}_r(1)\mathbf{A}_r(K)I_{4r+8}(\pi(1 + K^2)). \quad (\text{GS.18})$$

The right side is a finite positive expression and tends to zero as $J \rightarrow \infty$. In particular the infinite version of (GS.15) is absolutely convergent and equals μ_{2r} .

Proof. For $x \geq 1$, integers $a \geq 1$ and $p \geq 0$, write $n = a + j$. The inequality $(a + j)^2 - a^2 = 2aj + j^2 \geq (2a + 1)j$ proves

$$\sum_{n \geq a} n^p e^{-\pi n^2 x^2} \leq e^{-\pi a^2 x^2} \sum_{j \geq 0} (a + j)^p \rho_a^j = e^{-\pi a^2 x^2} \mathbf{H}_p(a, \pi). \quad (\text{GS.19})$$

For the equality, expand $(a + j)^p$ and then use $j^v = \sum_b \left\{ \begin{smallmatrix} v \\ b \end{smallmatrix} \right\} (j)_b$, where $(j)_b = j(j-1) \cdots (j-b+1)$. This identity follows inductively from $j(j)_b = (j)_{b+1} + b(j)_b$ and the displayed Stirling recurrence. Differentiating the geometric series b times gives $\sum_{j \geq 0} (j)_b \rho^j = b! \rho^b / (1 - \rho)^{b+1}$, which proves the last equality in (GS.19).

Since $x^{2b} \leq x^{2r+4}$ for every original exponent $b \leq r + 2$, (GS.19) gives the three explicit inequalities

$$|H_r(x)|, |H_{r,J}(x)| \leq \mathbf{A}_r(1)x^{2r+4}e^{-\pi x^2},$$

$$|H_r(x) - H_{r,J}(x)| \leq \mathbf{A}_r(K)x^{2r+4}e^{-\pi K^2 x^2}.$$

The same estimate for the sum of absolute values of all summands proves absolute integrability of the expanded double square. In particular termwise integration in the infinite formula is legitimate. Now use $|H_r^2 - H_{r,J}^2| \leq |H_r - H_{r,J}|(|H_r| + |H_{r,J}|)$ and the endpoint factor 2 in (GS.11). The resulting constant is exactly 4, the power is $4r + 8$, and integration gives (GS.18).

For fixed r , the denominators in (GS.16) are bounded away from zero uniformly for integers $a \geq 1$; thus $\mathbf{H}_p(a, \pi)$ is at most a constant times $(1 + a)^p$, and $\mathbf{A}_r(a)$ has at most polynomial growth. For $C \geq \pi$,

$$I_p(C) \leq e^{-C} \int_1^\infty x^p e^{-\pi(x^2-1)} dx.$$

The last integral is finite. Hence the factor with $C = \pi(1 + K^2)$ dominates that polynomial growth and forces $E_{r,J} \rightarrow 0$, proving the remaining assertions. $\square$

In the two new cases the coefficients in (GS.17) are completely specified by

$$\begin{aligned} A_2(a) &= 75\pi H_2 + 330\pi^2 H_4 + 224\pi^3 H_6 + 32\pi^4 H_8, \\ A_3(a) &= \frac{375}{2}\pi H_2 + 1635\pi^2 H_4 + 2116\pi^3 H_6 + 720\pi^4 H_8 + 64\pi^5 H_{10}, \end{aligned} \quad (\text{GS.20})$$

where each H_p on the right means $H_p(a, \pi)$. These absolute coefficients enter the error estimate only. Every original negative coefficient stays in the exact square (GS.15).

4 The coefficient morphism at the actual gamma reference

For $\lambda > 0$ put

$$r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda), \quad \alpha = 2\lambda. \quad (\text{GS.21})$$

We prove the constants needed for the coefficient map. The beta integral follows from the product of two gamma integrals by the substitution $v = x_1 + x_2$, $u = x_1/(x_1 + x_2)$ with Jacobian v . After $u/(1-u) = e^x$ it reads

$$\int_{\mathbb{R}} (2 \cosh(x/2))^{-2\lambda} e^{ivx} dx = \frac{|\Gamma(\lambda + iv)|^2}{\Gamma(2\lambda)}. \quad (\text{GS.22})$$

The integrand before transformation and its first two derivatives decay exponentially at both ends. Two integrations by parts therefore give an integrable Fourier transform, so Fourier inversion applies with the same Gaussian justification as above. Using its evenness and substituting $t = 2v$ gives

$$\int_{\mathbb{R}} e^{iyt} r_\lambda(t) dt = c_\lambda (\cosh y)^{-2\lambda}. \quad (\text{GS.23})$$

The factor 2 from $t = 2v$ is present in c_λ . The function $(2 \cosh(x/2))^{-2\lambda}$ is holomorphic on $|\Im x| < \pi$ with the branch positive on the real line and decays exponentially on each closed narrower strip. Shifting the contour in (GS.22) up for $v > 0$ and down for $v < 0$ proves $|\Gamma(\lambda + iv)|^2 \leq C_c e^{-c|v|}$ for each $0 < c < \pi$. Thus the density in (GS.21) has exponential moments for every exponent of absolute value less than $\pi/2$. Analytic continuation of (GS.23) yields

$$\int e^{\theta t} r_\lambda(t) dt = c_\lambda (\cos \theta)^{-\alpha} \quad (|\theta| < \pi/2). \quad (\text{GS.24})$$

Define the monic polynomials by the local generating function

$$F_\lambda(z, t) = (1 - iz)^{-\lambda+it/2} (1 + iz)^{-\lambda-it/2} = \sum_{n \geq 0} \frac{b_n^{(\lambda)}(t)}{n!} z^n, \quad (\text{GS.25})$$

$$b_0 = 1, \quad b_1 = t, \quad b_{n+1} = tb_n - n(n + \alpha - 1)b_{n-1}. \quad (\text{GS.26})$$

The logarithms in (GS.25) vanish at zero. For real z close to zero, $F_\lambda(z, t) = (1 + z^2)^{-\lambda} e^{t \arctan z}$. Equation (GS.24) and $\cos(\arctan z + \arctan v) = (1 - zv)/\sqrt{(1 + z^2)(1 + v^2)}$ give

$$\int F_\lambda(z, t) F_\lambda(v, t) r_\lambda(t) dt = c_\lambda (1 - zv)^{-\alpha}. \quad (\text{GS.27})$$

Exponential integrability permits every derivative at $(0, 0)$ under the integral. Coefficient comparison proves

$$\int b_n^{(\lambda)}(t) b_j^{(\lambda)}(t) r_\lambda(t) dt = c_\lambda n! (\alpha)_n \delta_{nj}. \quad (\text{GS.28})$$

The highest coefficient of (GS.25) is $t^n/n!$, so these polynomials are monic. Finally differentiating (GS.25) gives $(1 + z^2)\partial_z F_\lambda = (t - \alpha z)F_\lambda$; coefficient comparison proves (GS.26), including all its constants.

The actual amplitude linking the two measures is

$$A_{1,\lambda}(t) = \frac{g(1/2 + it)}{\Gamma(\lambda + it/2)}, \quad B_{1,\lambda}(t) = |A_{1,\lambda}(t)|^2, \quad w(t) = B_{1,\lambda}(t)r_\lambda(t). \quad (\text{GS.29})$$

Gamma has no zeros and no poles at the displayed arguments. Here is a direct proof on their half-plane. For $\Re z > 0$, repeated integration by parts gives $B(z, n+1) = n!/[z(z+1)\cdots(z+n)]$. Substitution $u = v/n$ in the beta integral and $(1 - v/n)^n \leq e^{-v}$ prove $n^z B(z, n+1) \rightarrow \Gamma(z)$ by dominated convergence. The reciprocal finite products converge to

$$ze^{\gamma z} \prod_{j \geq 1} (1 + z/j)e^{-z/j}, \quad \gamma = \lim_{n \rightarrow \infty} \left(\sum_{j=1}^n j^{-1} - \log n \right).$$

Integral comparison proves existence of the real limit. The logarithm of the tail product converges absolutely because $\log(1 + z/j) - z/j = O(j^{-2})$, and every factor is nonzero on $\Re z > 0$. The product is therefore finite and nonzero; by the previous limit it is $1/\Gamma(z)$. The original gamma integral is holomorphic on this half-plane, which also excludes poles there. The phase-preserving source map is multiplication by $A_{1,\lambda}$ from $L^2(w dt)$ to $L^2(r_\lambda dt)$; the exact norm identity follows by (GS.29). It is onto: the zero set of g on the line is discrete because g is entire and $g(2) > 0$ by (GS.5). Division by the amplitude off that null set gives the inverse and the same norm equality.

For completeness the full source phase at the chosen parameter is

$$A_{1,13/4}(t) = \frac{-\pi^{-1/4}(t^2 + 1/4)e^{-it \log \pi/2} \zeta(1/2 + it)}{(1/4 + it/2)(5/4 + it/2)(9/4 + it/2)}. \quad (\text{GS.30})$$

It follows from (GS.5), $s(s-1) = -(t^2 + 1/4)$, and three gamma recurrences. The elementary identity $\zeta(s) = s/(s-1) - s \int_1^\infty \{x\}x^{-s-1} dx$ first follows by summation by parts for $\Re s > 1$ and extends to $\Re s > 0$, $s \neq 1$, because its integral converges absolutely and locally uniformly there. On the line in (GS.30) it proves $|\zeta(s)| \leq 1 + 2\sqrt{t^2 + 1/4} \leq 4\sqrt{t^2 + 1/4}$. The squared modulus of the denominator in (GS.30) is $4^{-3}(t^2 + 1/4)(t^2 + 25/4)(t^2 + 81/4)$, so

$$0 \leq B_{1,13/4}(t) \leq \frac{1024}{\sqrt{\pi}} \frac{(t^2 + 1/4)^2}{(t^2 + 25/4)(t^2 + 81/4)} \leq \frac{1024}{\sqrt{\pi}}. \quad (\text{GS.31})$$

The last denominator exceeds the numerator by $26t^2 + 253/2 > 0$ after expansion. Thus $B_{1,13/4}$ belongs to $L^2(r_{13/4} dt)$, and its gamma coefficients have their literal Hilbert-space type.

Write $c = c_{13/4}$ and $c_j = c_{1,j}$. Equations (GS.28)–(GS.29) give the linear coefficient map on the moment functional:

$$c_j = \frac{\int b_j^{(13/4)}(t)w(t) dt}{c j!(13/2)_j}, \quad c = 2^{-11/2} \Gamma(13/2) = \frac{10395\sqrt{\pi}}{2048\sqrt{2}}. \quad (\text{GS.32})$$

The last identity follows by six gamma recurrences from $\Gamma(1/2) = \sqrt{\pi}$. At this exact parameter the complete polynomials needed are

$$\begin{aligned} b_0 &= 1, & b_1 &= t, & b_2 &= t^2 - \frac{13}{2}, \\ b_3 &= t^3 - \frac{43}{2}t, & b_4 &= t^4 - 47t^2 + \frac{663}{4}, \\ b_5 &= t^5 - 85t^3 + \frac{3931}{4}t, \\ b_6 &= t^6 - \frac{275}{2}t^4 + \frac{13801}{4}t^2 - \frac{69615}{8}. \end{aligned} \quad (\text{GS.33})$$

Substituting successively into (GS.26) proves every line. Evenness gives $c_1 = c_3 = c_5 = 0$, and the exact even coefficient map becomes

$$\begin{aligned} c_0 &= \mu_0/c, \\ c_2 &= (2\mu_2 - 13\mu_0)/(195c), \\ c_4 &= (4\mu_4 - 188\mu_2 + 663\mu_0)/(377910c), \\ c_6 &= (8\mu_6 - 1100\mu_4 + 27602\mu_2 - 69615\mu_0)/(2737957950c). \end{aligned} \tag{GS.34}$$

Here $2!(13/2)_2 = 195/2$, $4!(13/2)_4 = 188955/2$, and $6!(13/2)_6 = 1368978975/4$. These factors show directly how every denominator in (GS.34) arises. The invertible triangular moment map has the complete inverse

$$\begin{aligned} \mu_0 &= cc_0, \\ \mu_2 &= (195cc_2 + 13\mu_0)/2, \\ \mu_4 &= (377910cc_4 + 188\mu_2 - 663\mu_0)/4, \\ \mu_6 &= (2737957950cc_6 + 1100\mu_4 - 27602\mu_2 + 69615\mu_0)/8. \end{aligned} \tag{GS.35}$$

Substitution in (GS.34) proves that the displayed maps are mutual inverses. In particular the mass in (GS.7) remains μ_0 , while the reference mass in (GS.21) remains c .

5 Certified infinite quantities and exact rational error transport

The companion checker evaluates (GS.15) for $r = 0, 1, 2, 3$ and $J = 8$ in Arb real balls at 256-bit precision, with one arithmetic thread and python-flint 0.9.0. All polynomial input is exact rational arithmetic; π , exponentials, powers and gamma functions are enclosed by Arb. The upper-incomplete-gamma call is `C.gamma_upper(alpha)`, with argument C and order α , precisely as in (GS.13). Each finite ball is enlarged by a symmetric ball with radius the upward-rounded upper endpoint of (GS.18). It therefore encloses the infinite quantity. The computed tails satisfy the rational inequalities

$$E_{0,8} < 7 \cdot 10^{-107}, \quad E_{1,8} < 4 \cdot 10^{-103}, \quad E_{2,8} < 2 \cdot 10^{-99}, \quad E_{3,8} < 14 \cdot 10^{-96}. \tag{GS.36}$$

Each inequality is tested as a strict rigorous ball comparison.

The resulting exact rational decimal intervals are:

$$\begin{aligned}
\mu_0 &\in [1.2790072478464851404795335922671932744916, \\
&\quad 1.2790072478464851404795335922671932744919], \\
\mu_2 &\in [13.0555493025705584353926846581231936824343, \\
&\quad 13.0555493025705584353926846581231936824346], \\
\mu_4 &\in [383.2743417173642425544072374791308324300288, \\
&\quad 383.2743417173642425544072374791308324300291], \\
\mu_6 &\in [17946.5273016052076235667646919834204772848931, \\
&\quad 17946.5273016052076235667646919834204772848934], \\
c &\in [6.3614260045872192926298535487331092297380, \\
&\quad 6.3614260045872192926298535487331092297383], \\
c_0 &\in [0.2010566886927858658099161632202187798614, \\
&\quad 0.2010566886927858658099161632202187798617], \\
c_2 &\in [0.0076454429994864011881338514807939967436, \\
&\quad 0.0076454429994864011881338514807939967439], \\
c_4 &\in [-0.0000305169208037620437467248395269085390, \\
&\quad -0.0000305169208037620437467248395269085387], \\
c_6 &\in [-0.0000003851554876709816618220743079695913, \\
&\quad -0.0000003851554876709816618220743079695910].
\end{aligned}$$

Each interval has width $3 \cdot 10^{-40}$. The coefficient balls are computed directly from the infinite-moment balls in (GS.34), before their conversion to the displayed decimal intervals. Interval dependency can enlarge enclosures; it cannot invalidate inclusion. The signs $c_4 < 0$ and $c_6 < 0$ follow from their negative upper endpoints and are preserved.

Here is the exact outward conversion used by the checker. If the resulting Arb ball is Q , let ℓ and u be its outward dyadic endpoints and set $d = 40$,

$$L_Q = 10^{-d}(\lfloor 10^d \ell \rfloor - 1), \quad U_Q = 10^{-d}(\lceil 10^d u \rceil + 1). \quad (\text{GS.37})$$

The checker then verifies $Q > L_Q$ and $Q < U_Q$ by rigorous ball comparison. The receipt stores L_Q, U_Q both as decimal strings and exact rational strings, and stores the dyadic endpoints as exact rationals. Thus every display can be checked with integer arithmetic after ball generation. The same receipt records the tail's exact upward dyadic rational.

The executable finite checks additionally verify all exact rational polynomial lists in (GS.12) and (GS.33) and the four exact rational denominator factors following (GS.34). They compare the ball evaluations of the polynomial and integer numerator forms in (GS.34) for overlap, and similarly compare the endpoint-containing recurrence (GS.14) at each argument pair used. These checks detect convention mistakes; the analytic certificate is the ball enclosure enlarged by the proved infinite tail, independently of any truncation agreement.

Four fault modes inject specific false mathematical assertions: discarding the inversion factor two preserves μ_4 ; assigning reference mass one preserves c_0 ; c_4 is nonnegative; and the cutoff-zero sum with zero error encloses the full positive seed mass. The already certified disjoint intervals reject each assertion. These checks use explicit conditionals and a nonzero exit status; Python optimization cannot remove them. Normal and optimized runs produce the same mathematical results and each fault produces one recorded rejection.

The source note's two floating truncations remain its original numerical evidence. The intervals above independently certify their displayed values through the original theta series and (GS.18). The correctness contract for the elementary ball operations and outward endpoints is documented in the python-flint Arb interface. No proof of that arithmetic library's implementation is asserted.

6 Exact tensor coefficient interface through degree six

The map from the certified one-factor data to every finite tensor degree can also be stated without changing the mass or sum variable. Let $k \geq 1$, $u = t_1 + \cdots + t_k$, and $m_k = w^{*k}$. Fubini gives $\int m_k = \mu_0^k$, and exponential integrability proves all the moments below are finite. Multiplying the generating functions (GS.25) gives the finite polynomial identity

$$b_n^{(k\lambda)}(u) = \sum_{n_1 + \cdots + n_k = n} \frac{n!}{n_1! \cdots n_k!} \prod_{j=1}^k b_{n_j}^{(\lambda)}(t_j). \quad (\text{GS.38})$$

Its proof is literal multiplication: the exponents of $1 \mp iz$ add to $-k\lambda \pm iu/2$, and the coefficient of degree n has only finitely many terms.

Use $\lambda = 13/4$, $\alpha = 13/2$, $a_j = (\alpha)_j c_j$, and the formal coefficient series $\mathcal{A}(z) = \sum_{j \geq 0} a_j z^j$. For any coefficient appearing below only finitely many c_j are needed. Integration of (GS.38), using (GS.32) separately in each factor, proves

$$\int b_n^{(k\lambda)}(u) m_k(u) du = c^k n! d_{k,n}, \quad d_{k,n} = [z^n] \mathcal{A}(z)^k. \quad (\text{GS.39})$$

No interchange of an infinite formal series with an integral is required. Evenness gives the complete first four nonzero expressions

$$\begin{aligned} d_{k,0} &= c_0^k, \\ d_{k,2} &= k c_0^{k-1} a_2, \\ d_{k,4} &= k c_0^{k-1} a_4 + \binom{k}{2} c_0^{k-2} a_2^2, \\ d_{k,6} &= k c_0^{k-1} a_6 + k(k-1) c_0^{k-2} a_2 a_4 + \binom{k}{3} c_0^{k-3} a_2^3. \end{aligned} \quad (\text{GS.40})$$

For degree six the partitions into even positive degrees are 6, 4+2 and 2+2+2; their distinct-factor choices have respectively k , $k(k-1)$ and $\binom{k}{3}$ possibilities. This proves all three coefficients, including the mixed term and its sign. The analogous enumeration proves degree two and four. Since $c_0 > 0$, even the displayed negative powers at $k = 1, 2$ are defined; their accompanying zero combinatorial factors give the correct absent terms.

Equations (GS.34)–(GS.35), (GS.39)–(GS.40) are the exact map from the certified original moments to these all- k low-degree arithmetic gamma moments, retaining c^k and the coordinate u . For example their degree-zero identity is $c^k d_{k,0} = \mu_0^k$. They give interval evaluations at every specified finite integer k by rational and ball arithmetic with the already enclosed inputs. They assert no uniform estimate when polynomial degree grows. For this actual seed $h = 1$, the finite arithmetic quotient is $\mathbb{C}[S]/(1) = 0$: the quotient map sends every polynomial to zero because every polynomial lies in the ideal (1). The analytic source measure and all the coefficients above remain nonzero before that quotient. No nonzero packet quotient is inferred from these seed calculations.